Jay Darshan Vakil

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Research Interests

Embodied Artificial Intelligence, Data-Driven Robot Learning, Mobile Manipulation

Education

Doctor of Philosophy in Computer Science, *University of Colorado* | Boulder, CO *Aug 2024 - Present*

- **Advisors:** Alessandro Roncone

Bachelor of Science in Electrical Engineering, *University of Washington* | Seattle, WA *Sept 2018 - Mar 2022*

- Minor in Mathematics and Computer Science
- **Awards:** Dean's List
- **Courses:** Object-Oriented programming with Java, Software Engineering, Data structures and algorithm, Linear/discrete signal processing, Microprocessor system design, AC/DC circuit analysis, Digital circuit analysis, Electrodynamics, Transistors and amplifiers, Electrical testing, Biomedical instrumentation.

Research Experience

MyoLab.AI, *Research Scientist Intern* | New York City, NY *Feb 2025 - May 2025*

- **Research Focus:** Multimodal agent LLM based planning, Diffusion-based human-motion synthesis, Actor count control, Human motion analysis and control, Temporal coherence, Embodied AI, Text-to-Motion motion generation.
- Built a two-level architecture with high-level LLM semantic planner + low-level mixture of experts controllers.
- Diffusion priors on human-motion manifolds; temporally coherent generative modeling; latent-space interpolation/blending/transition operators
- Controllability, disentanglement, and identifiability of motion/style/shape factors; actor-count conditioning; scene composition control
- Interactive, human-in-the-loop synthesis; low-latency closed-loop editing; camera/playback trajectory optimization
- Generalization and robustness (few-shot, OOD scenes); data curation and prompt-to-action alignment; ablations and user studies

Facebook AI Research, *Robotics Engineer* | Pittsburgh, PA and Fremont, CA *Apr 2022 - July 2024*

- **Research Focus:** Visual imitation learning for robot manipulation, Offline reinforcement learning, Mobile manipulation, Visual language models, Instance exploration and navigation
- Designed and constructed a distributed robotic arm cluster specifically tailored for large-scale experimentation and data collection. This cluster served as the hardware backbone for numerous impactful research projects.
- Researched a generalized universal agent proficient in solving challenging manipulation tasks within a kitchen-top environment. Our approach efficiently addressed over 38 tasks exhibiting 12 skills. By incorporating world priors from existing foundational models, we augmented the dataset, enhancing the agent's ability to generalize across different scenes. Outperformed prior methods by over 40%.
- Curated and released RoboSet, a scalable multi-task real-world robotic dataset for generalized offline learning.
- Conducted a comprehensive empirical investigation on leveraging pre-trained visual representations for training real-world task execution policies, uncovering insights about the transferability from simulation to real-world environments, achieving groundbreaking results in indoor navigation, and showcasing the advantages of data augmentation and fine-tuning for real-world performance improvement. Evaluated the performance of five different PVRs across two policy-learning paradigms (visual imitation and reinforcement learning) and on three distinct robotic platforms.
- Developed a mobile manipulation agent capable of autonomously planning and exploring environments, while simultaneously recording instances of objects of interest in a 3D map. Utilizing a visual-language model and 3D perception backbone, the agent can accurately identify and localize objects and their corresponding instances based on open-vocabulary requests. Subsequently, the robot can navigate to the specified object, pick it up, and then transport it to a target location for placement.

Publications

Learning Whole-Body Representations for Collision-Aware Manipulation

Jay Vakil, Amine Boubrit, Artemis Shaw, Ekshan Raj Verma, Anuj Pasricha, Ava Abderezaei, Carson Kohlbrenner, Nikolaus Correll, Alessandro Roncone

- Learning from whole-body proximity sensing for collision-aware manipulation in constrained and confined spaces.
- In submission

Dynamics-Compliant Trajectory Diffusion for Super-Nominal Payload Manipulation

Anuj Pasricha, Joewie Koh, **Jay Vakil**, Alessandro Roncone

- Payload conditioned diffusion model that denoises joint angles, velocities, and torques to generate dynamic motions at 3x nominal payload.
- Accepted at **CoRL 2025**
- Website Paper

Towards Open-World Mobile Manipulation in Homes: Lessons from the Neurips 2023 HomeRobot Open Vocabulary Mobile Manipulation Challenge

Sriram Yenamandra*, Arun Ramachandran*, Mukul Khanna*, Karmesh Yadav*, **Jay Vakil***, Andrew Melnik, Michael Büttner, Leon Harz, Lyon Brown, Gora Chand Nandi, Arjun PS, Gaurav Kumar Yadav, Rahul Kala, Robert Haschke, Yang Luo, Jinxin Zhu, Yansen Han, Bingyi Lu, Xuan Gu, Qinyuan Liu, Yaping Zhao, Qiting Ye, Chenxiao Dou, Yansong Chua, Volodymyr Kuzma, Vladyslav Humennyy, Ruslan Partsey, Jonathan Francis, Devendra Singh Chaplot, Gunjan Chhablani, Alexander Clegg, Theophile Gervet, Vidhi Jain, Ram Ramrakhya, Andrew Szot, Austin Wang, Tsung-Yen Yang, Aaron Edsinger, Charlie Kemp, Binit Shah, Zsolt Kira, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton

- Open Vocabulary Mobile Manipulation as a key benchmark task for robotics: finding any object in a novel environment and placing it on any receptacle surface within that environment.
- Website Paper

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration

- The largest robot learning dataset with 1 million+ real-world robot trajectories over 22 robot embodiments.
- Accepted at **International Conference on Robotics and Automation (ICRA) 2024 (Best Conference Paper Award)**
- Website Paper

OK-Robot: What Really Matters in Integrating Open-Knowledge Models for Robotics

Peiqi Liu*, Yaswanth Orru*, **Jay Vakil**, Chris Paxton, Nur Muhammad "Mahi" Shafiullah†, Lerrel Pinto†

- A zero-shot integrated framework for open vocabulary mobile manipulation tasks by leveraging visual-language models and navigation primitives.
- Accepted at **RSS 2024**, **Robotic Tasks and How to Specify Them? (RSS 2024 workshop)**, **Workshop on Semantic Reasoning and Goal Understanding in Robotics (RSS 2024)**
- Website Paper

RoboHive: A Unified Framework for Robot Learning

Vikash Kumar, Rutav Shah, Gaoyue Zhou, Vincent Moens, Vittorio Caggiano, **Jay Vakil**, Abhishek Gupta, and Aravind Rajeswaran

- A comprehensive platform for Robot Learning and Embodied AI research, offering unified task interfaces, diverse environments, and expert benchmarks for algorithmic research in various domains.
- Accepted at **NeurIPS 2023**
- Website Paper

RoboAgent: Towards Sample Efficient Robot Manipulation with Semantic Augmentations and Action Chunking

Homanga Bharadhwaj*, **Jay Vakil***, Mohit Sharma*, Abhinav Gupta, Shubham Tulsiani, and Vikash Kumar

- An efficient end-to-end pipeline for training universal agents with multi-task manipulation skills using semantic augmentations and compact action representations, achieving superior performance and adaptability in a low-data regime.
- Accepted at **ICRA 2024**, **OOD Workshop at CoRL 2023**, **Outstanding Presentation Award at Robot Learning Workshop, NeurIPS 2023**
- Website Paper

What do we learn from a large-scale study of pre-trained visual representations in sim and real environments?

Sneha Silwal*, Karmesh Yadav*, Tingfan Wu*, **Jay Vakil***, Arjun Majumdar*, Sergio Arnaud*, Claire Chen, Vincent-Pierre Berges, Dhruv Batra, Aravind Rajeswaran, Mrinal Kalakrishnan, Franziska Meier, and Oleksandr Maksymets

- Extensive research on using pre-trained visual representations (PVRs) for real-world task execution, involving multiple PVRs, policy-learning methods, and robots. Identified key insights, including the correlation between simulation and real-world performance and the success of PVRs in achieving zero-shot indoor navigation.
- Accepted at **ICRA 2024**, and **MoMa workshop ICRA 2024**
- Website Paper

Where are we in the search for an Artificial Visual Cortex for Embodied Intelligence?

Arjun Majumdar, Karmesh Yadav, Sergio Arnaud, Yecheng Jason Ma, Claire Chen, Sneha Silwal, Aryan Jain, Vincent-Pierre Berges, Tingfan Wu, **Jay Vakil**, Pieter Abbeel, Jitendra Malik, Dhruv Batra, Yixin Lin, Oleksandr Maksymets, Aravind Rajeswaran, Franziska Meier

- Conducted the largest study on pre-trained visual representations (PVRs) for Embodied AI, creating CORTEXBENCH with 17 tasks. Demonstrated that data scaling doesn't universally improve performance, but task-specific adaptation of VC-1 models achieved competitive or superior results on all benchmarks.
- Accepted at **NeurIPS 2023**
- Website Paper

SLAP: Spatial-Language Attention Policies

Priyam Parashar, Vidhi Jain, Xiaohan Zhang, **Jay Vakil**, Sam Powers, and Chris Paxton

- Address challenges in language-guided mobile manipulation, achieving an 80% success rate in real-world tasks with a single model, and a 30% improvement over prior work in challenging conditions with limited data, demonstrating robust execution of open-vocabulary task plans.
- Accepted at **CoRL 2023**
- Website Paper

Teaching Experience

University of Colorado, *Teaching assistant* | Boulder, CO *August 2024 - Present*
- Introduction to Robotics (CSCI 3302)
- Introduction to Artificial Intelligence (CSCI 3202)
- Computer systems (CSCI 2400)

Select media coverage

- **Carnegie Mellon University** Parenting a 3-Year-Old Robot
- **TechCrunch** Human toddlers are inspiring new approaches to robot learning
- **IEEE Spectrum** Video Friday: A 3-Year-Old Robot Your weekly selection of awesome robot videos
- **Hackaday** ROBOAGENT GETS ITS MT-ACT TOGETHER
- **TechBrief** Meet RoboAgent: Enabling Robots to Acquire Manipulation Abilities
- **Communications of the ACM** Parenting a Three-Year-Old Robot
- **Venture Beat** Meta's OK-Robot performs zero-shot pick-and-drop in unseen environments
- **MIT Technology Review** This robot can tidy a room without any help
- **Fox News** Stop loading the dishwasher; this robot aims to do all the cleanup for you
- **AI at Meta** Robots that learn from videos of human activities and simulated interactions
- **Rocking Robots** 'RoboAgent' Modeled After Human Infants' Learning Techniques
- **Mark Tech Post** This AI Research Presents RoboHive: A Comprehensive Software Platform and Ecosystem for Research in the Field of Robot Learning and Embodied Artificial Intelligence
- **StudyFinds** Meta AI researchers are training robots to learn like 3-year-olds

Academic service

Reviewer

- Conference on Robot Learning 2026 (CoRL)
- Human-Robot Interaction 2027 (HRI 2027)
- Neural Information Processing Systems (NeurIPS 2026)
- Intelligent Robots & Systems 2026 (IROS 2026)
- International Conference on Learning Representations 2026 (ICLR 2026)
- IEEE Robotics and Automation Letters (RA-L)
- Robotics: Science and Systems, RobotEvaluation Workshop (RSS 2025)
- International Conference on Machine Learning, Building Physically Plausible World Models (ICML 2025)
- IEEE International Conference on Robotics and Automation 2025 (ICRA 2025)
- IEEE International Conference on Robotics and Automation 2024 (ICRA 2024)
- Conference on Robot Learning 2024 (CoRL 2024)

Skills

Programming	Python, C/C++, C#, Java, CUDA, Matlab, Bash, LaTeX, Vim, Linux, Tensorflow, Pytorch, Docker, OpenCV, AWS S3, Git
Robotics/Hardware	ROS 1/2, MuJoCo, Arduino, Raspberry PI, Sensor Interfacing, Signal processing, Digital/Analog circuit design and simulation, Microprocessors, Embedded systems, Circuit testing, Transistor-level design, CAD, Electrical circuit testing, Isaac Sim, Isaac Lab